Dynamic drivers

palm group

Institute of Meteorology and Climatology, Leibniz Universität Hannover

Dynamic drivers

Dynamic driver

- Dynamic driver: all data which is variable in <u>time</u>
  - initialization data for atmosphere (wind, potential temperature, humidity) and soil state (temperature, moisture)
  - geostrophic wind
  - boundary data for offline nesting (at all lateral boundaries of the domain)
- Initialization data can be provided in different level of detail (lod):
  - LOD 1: profiles of, e.g., potential temperature (mapped to all locations (x,y))
  - LOD 2: 3D initialization data of, e.g., potential temperature for each grid point (x,y,z)
- Dynamic forcing data:
  - LOD 1: domain-averaged vertical profiles + nudging
  - LOD 2: time-dependent 2D slices at each lateral boundary of the domain
- See full description of the PIDS: http://palm-model.org/trac/wiki/doc/app/iofiles/pids
- **IMPORTANT NOTE:** LOD 1 has not been fully implemented yet!

  → LOD 2 is the default at the moment

Dynamic drivers

Dynamic driver

- **When to use a dynamic driver?**
  - General: to nest PALM into a mesoscale model (offline)
    - Real-case studies – recreating events or measuring campaigns
    - To consider synoptic-scale / mesoscale effects and changes (simulations where large-scale winds and thermodynamics are not constant)
  - Alternatively: to feed in data from a precursor simulation at the left model boundary

Dynamic drivers

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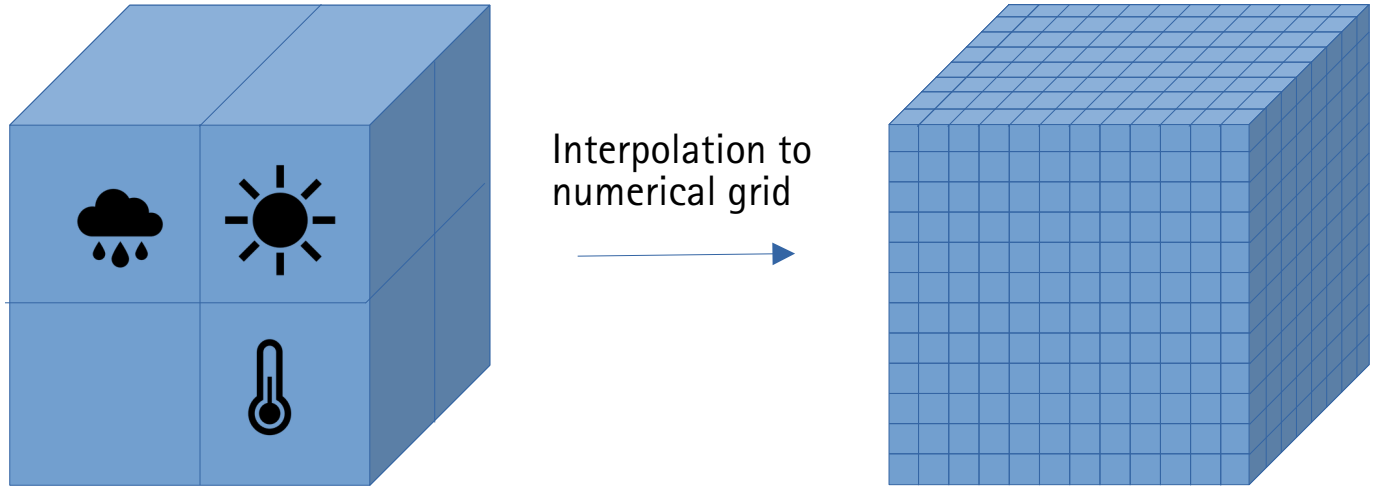
Dynamic drivers

Dynamic driver

- LOD 2 model initialization:

  Data must be interpolated to PALM's grid before it is compiled into the dynamic driver

Weather data: e.g. COSMO, ICON, WRF



Dynamic driver

- LOD 2 model forcing:

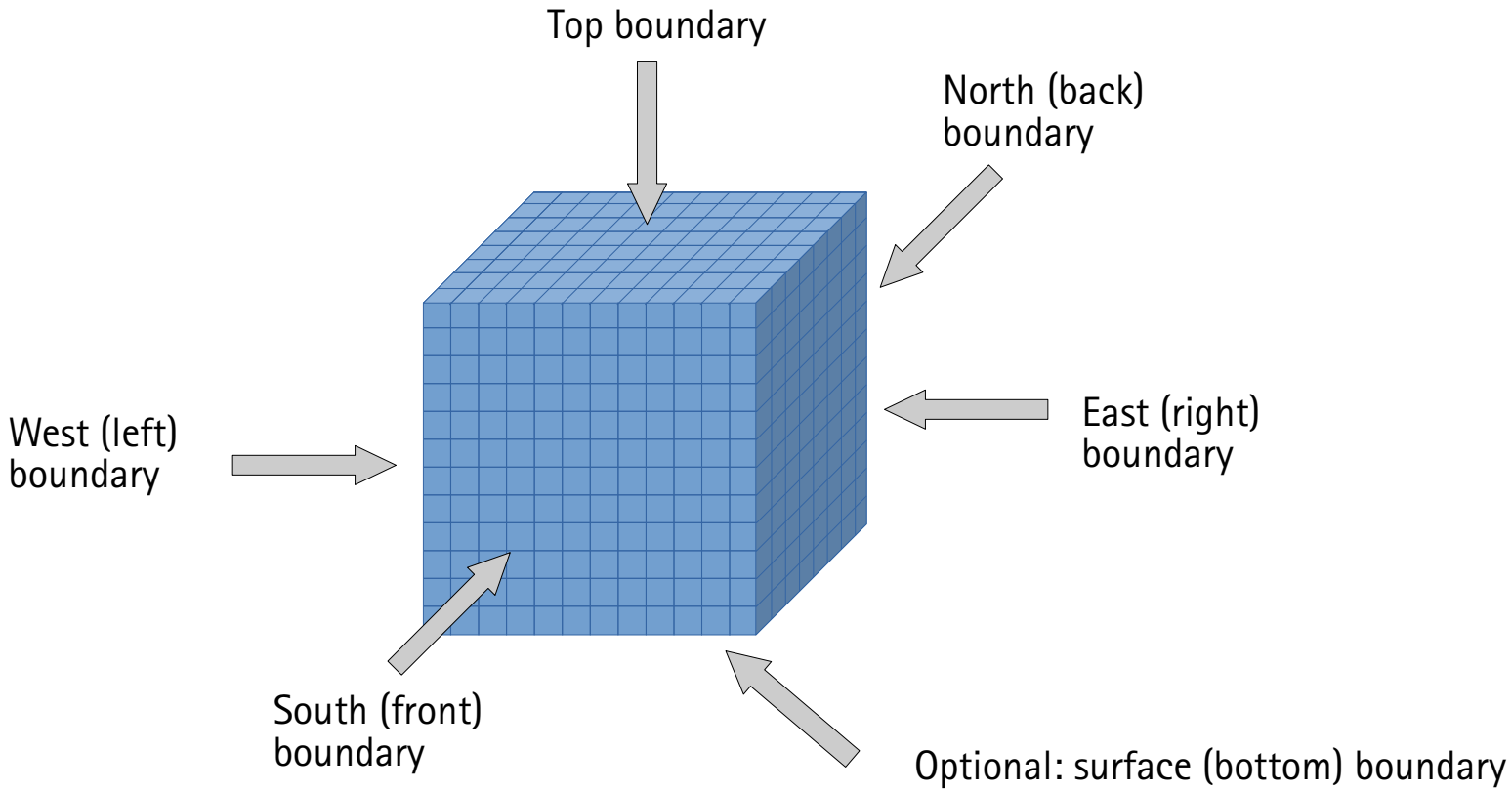
  Data must be provided at the lateral boundaries

Dynamic drivers

Dynamic driver

- LOD 2 model <u>forcing</u>:

  Data must be provided at the 5+1 lateral boundaries



Dynamic drivers

Dynamic driver – Example

- **Model initialization LOD 0**

  Steering via config file (p3d):

```
pt_vertical_gradient        = 0.0,      1.0,
pt_vertical_gradient_level = 0.0,   1400.0,
q_vertical_gradient         = 0.0,   -0.004,    0.0,
q_vertical_gradient_level   = 0.0,   1400.0, 1500.0,

uv_heights = 0.0, 25.0, 50.0, 100.0, 200.0, 400.0, 800.0, 1600.0, ...
u_profile  = 0.0, 2.0, 3.0, 4.2, 5.3, 6.4, 7.5, 8.5, ...
v_profile  = 0.0, 1.0, 1.5, 2.1, 2.7, 3.3, 3.9, 4.5, ...
```

- **Model initialization LOD 1 or LOD 2**

  → provide meteorological dat in NetCDF-format in the INPUT folder:
  `.../JOBS/<job_identifier>/INPUT/<job_identifier>`**`_dynamic`**

Dynamic drivers

Dynamic driver – Example

- Model initialization LOD 1:

```
init_atmosphere_pt(z)     ! Initial temperature profile
init_atmosphere_qv(z)     ! Initial humidity profile
init_atmosphere_u(z)      ! Initial wind profile (u)
…
init_soil_t(z)            ! Initial soil temperature
init_soil_m(z)            ! Initial soil moisture
```

- Model initialization LOD 2:

```
init_atmosphere_pt(z,y,x)  ! Initial temperature field
init_atmosphere_qv(z,y,x)  ! Initial humidity field
init_atmosphere_u(z,y,x)   ! Initial wind field (u)
…
init_soil_t(z,y,x)         ! Initial soil temperature field
init_soil_m(z,y,x)         ! Initial soil moisture field
```

Dynamic drivers

Dynamic driver – Example

- Model forcing LOD 1:

```
ls_forcing_ug(time,z)          ! Geostrophic wind (u)
ls_forcing_vg(time,z)          ! Geostrophic wind (v)
…
ls_forcing_adv_lpt(time,z)     ! Advection of Temperature
…
```

- Model forcing LOD 2:

```
ls_forcing_left_pt(time,z,y,x) ! Temperature at left boundary
ls_forcing_right_pt(time,z,y,x)! Temperature at right boundary
...
```

- Full list of parameters:

https://palm.muk.uni-hannover.de/trac/wiki/doc/app/iofiles/pids/dynamic

Dynamic drivers

Dynamic driver - steering

- To enable initialization with dynamic driver, set

```
&initialization_parameters
…
            initializing_actions = 'read_from_file'
```

- To enable offline nesting with dynamic driver, set NAMELIST:

```
&nesting_offl_parameters  /
```

Dynamic drivers

Dynamic driver – driver generation

- You can set up the dynamic driver according to the PIDS yourself (but that might take a lot of effort…)

- Current tools for creating dynamic drivers:
  - inifor:
    COSMO-D2 data (only for dates before February 2021)
    https://palm.muk.uni-hannover.de/trac/wiki/doc/app/iofiles/inifor
  - PROMET:
    ICON-D2/ICON-ART and WRF/WRF-Chem data
    https://gitlab.palm-model.org/dwd/promet
  - PALM-METEO:
    WRF, ICON, Aladin, Synthetic inputs (manually definable)
    https://github.com/PALM-tools/palmmeteo

Dynamic drivers

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inifor

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Geoscientific Model Development
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Mesoscale nesting interface of the PALM model system 6.0

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Revised: 2 June 2021 – Accepted: 13 July 2021 – Published: 3 September 2021

ary 2021)
oc/app/iofiles/inifor

https://gitlab.palm-model.org/dwd

- PALM-METEO:
  WRF, ICON, Aladin, Synthetic input
  https://github.com/PALM-tools/pal

https://doi.org/10.5194/egusphere-2025-4120
Preprint. Discussion started: 11 September 2025
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EGUsphere
Preprint repository

PALM-meteo

PALM-meteo 2.6: Processor of PALM meteorological input data

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Dynamic drivers

Dynamic driver – driver generation

- You find all tools in the PALM repository:

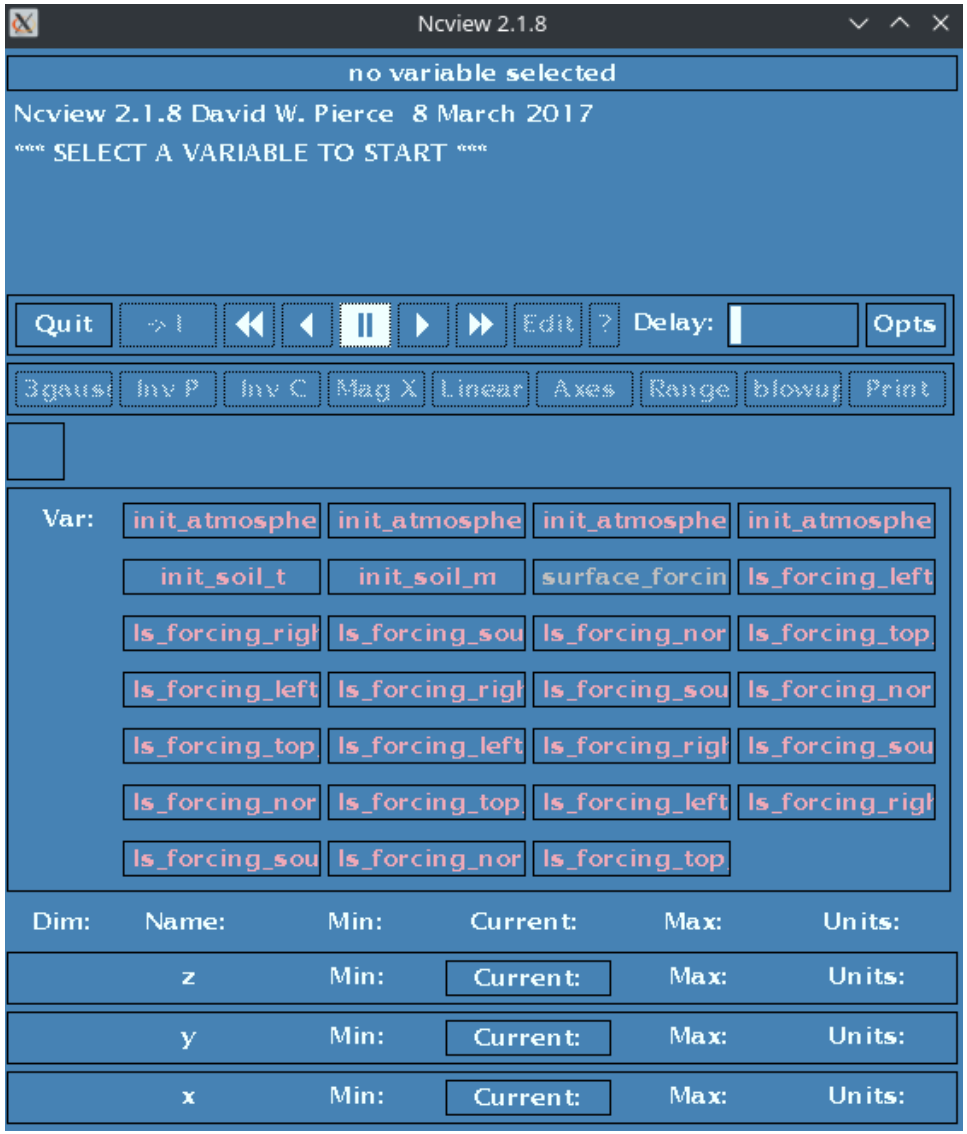
```
[...]/palm/[...]/packages/dynamic_driver/

inifor/
palm_meteo/
promet/
wrf_interface/ (old)
```

- Example dynamic driver **`promet_test_01_dynamic`** is available on seafile (Thursday - Lectures)

Dynamic drivers

Dynamic driver – example NetCDF



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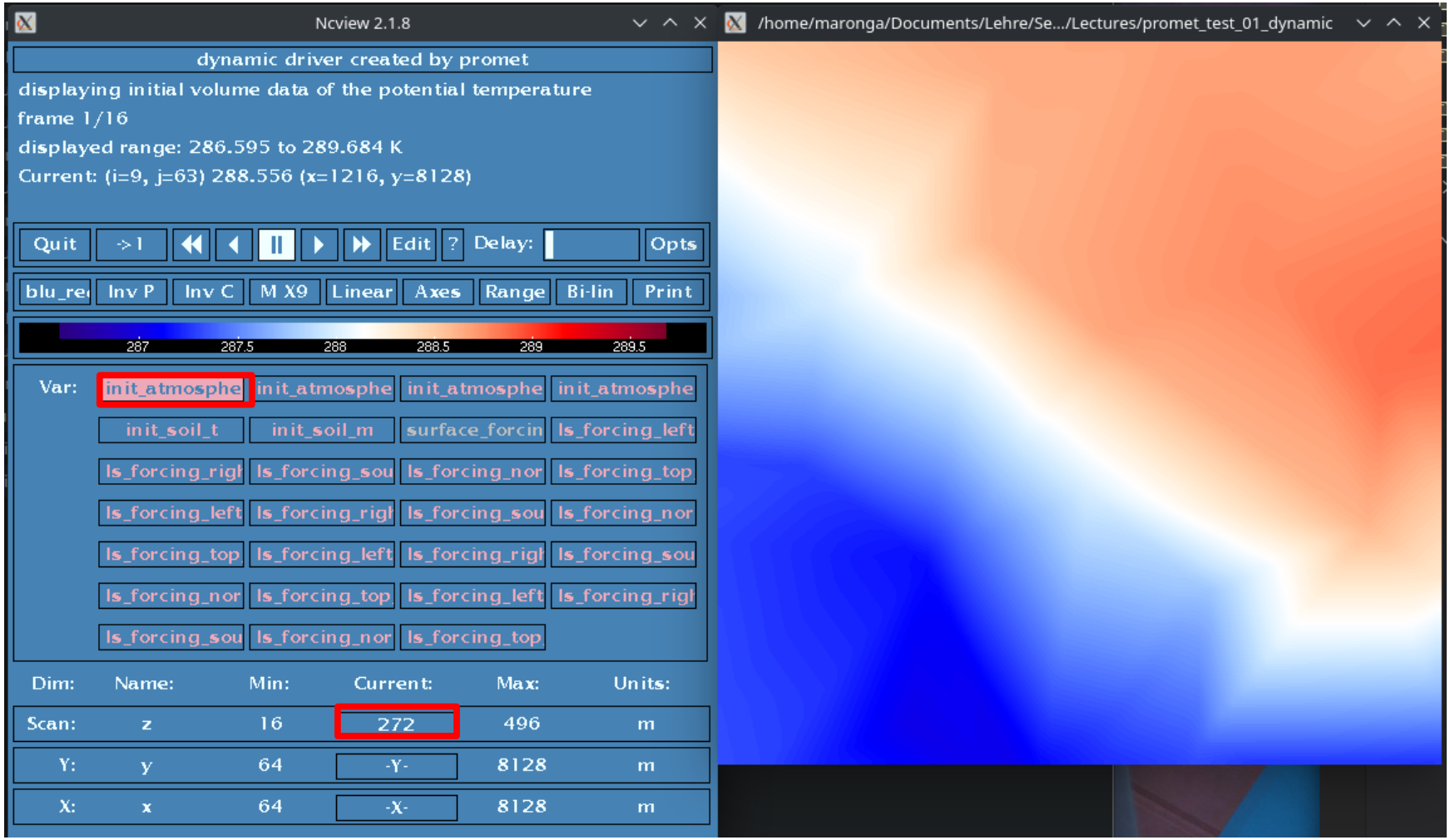

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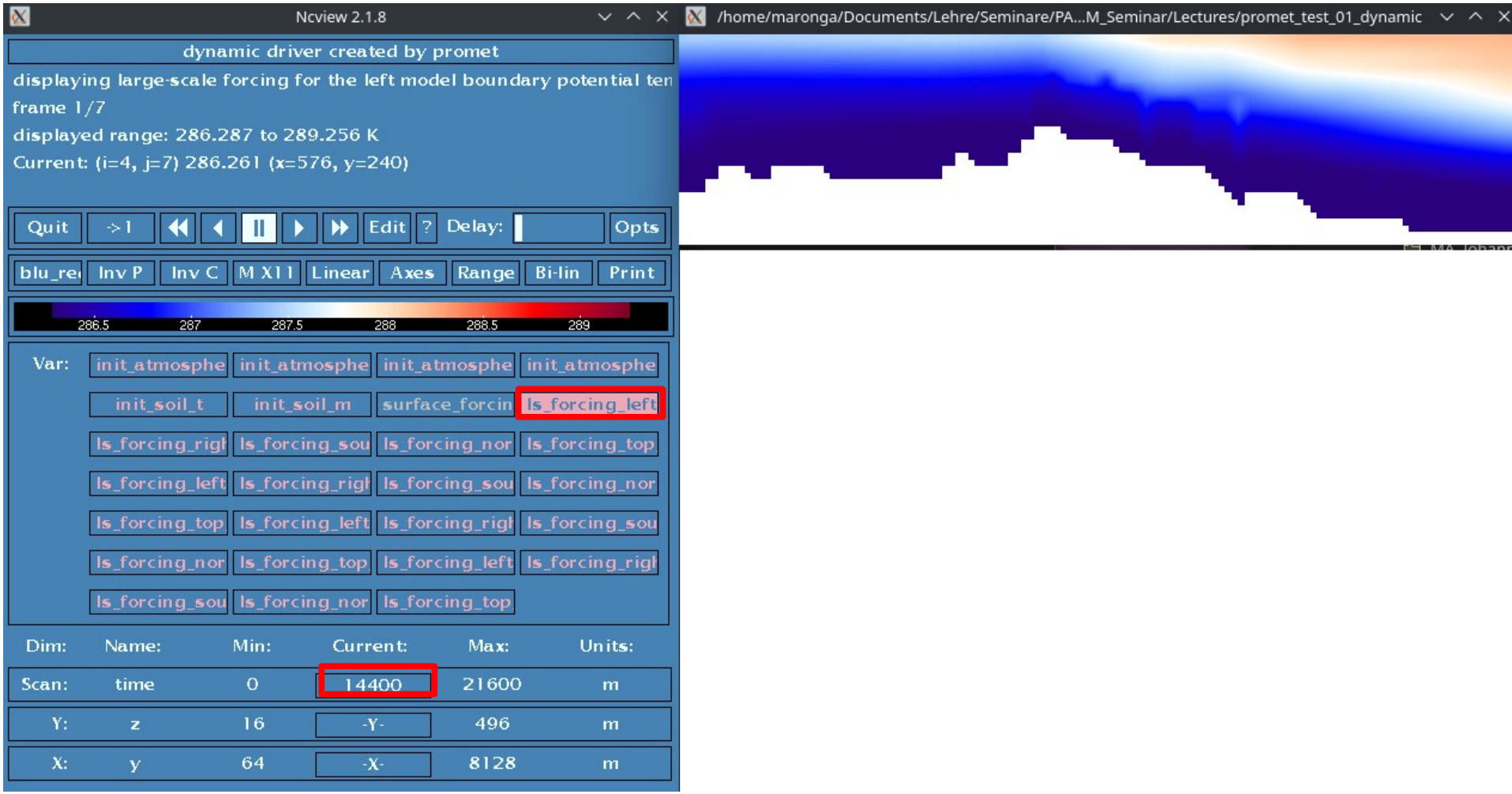

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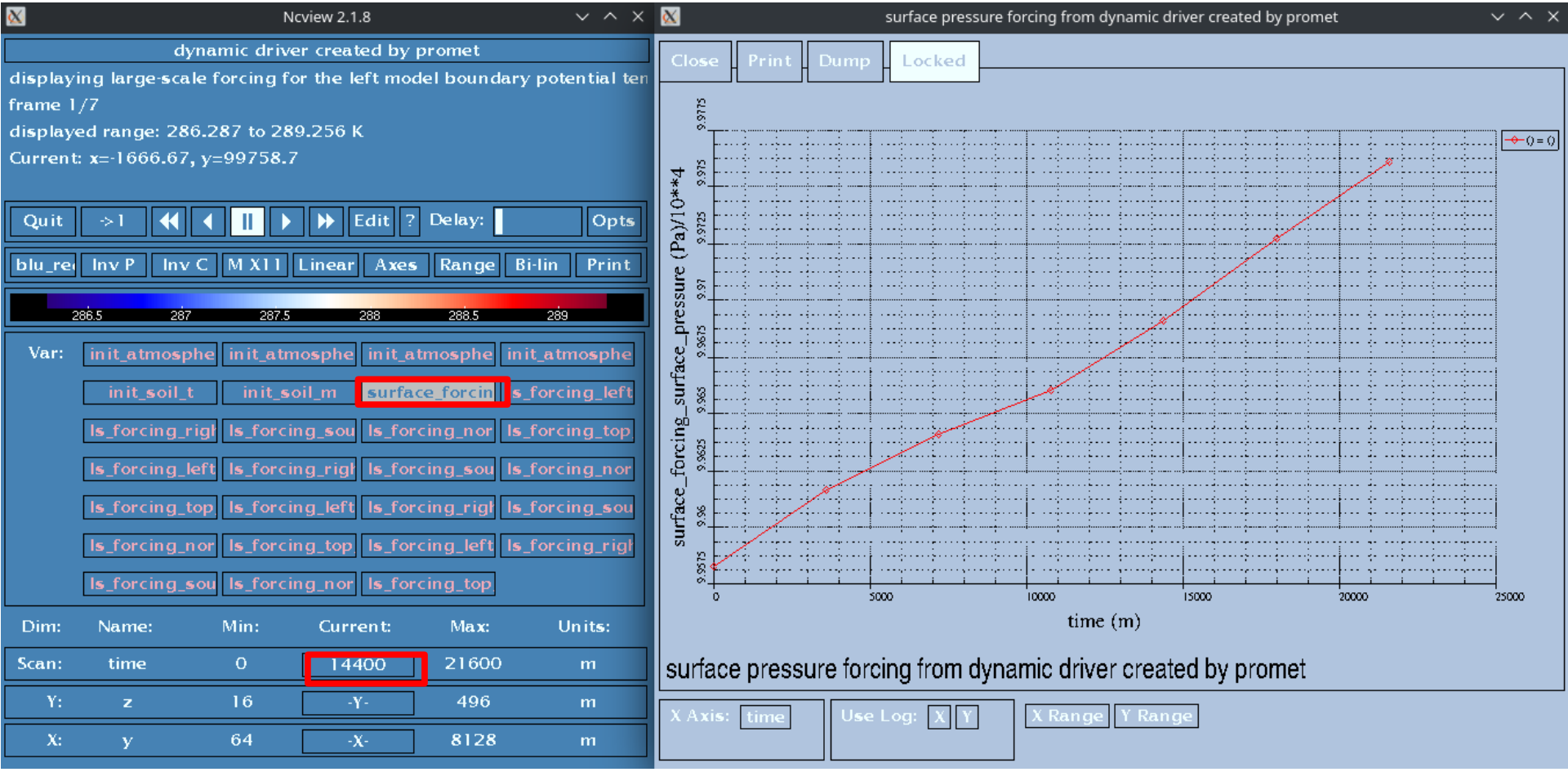

Dynamic drivers

Dynamic driver – current limitations & hints

- No universal pre-processing tool for all kind of mesoscale models
- We have not fully implemented (and tested!) offline nesting of cloudy air and precipitation
- In order to simulate situations with clouds, you can use the dynamic driver to prescribe radiative fluxes (see parameter `radiation_scheme = 'external'`)
- There is a pre-processing tool `pcycle` which converts PALM data from a precursor run into a dynamic driver (an alternative to turbulent inflow with recycling)
- LOD 1 is not fully implemented in PALM, although it is defined in the PIDS.
- |

PALM online:
https://palm.muk.uni-hannover.de

Our YouTube channel:
https://youtube.com/user/palmhannover